Experiment 18 - Experiment to understand KNN algorithm for a given data set.

Aim:

To understand the KNN algorithm for a given data set.

Program:

```
import numpy as np
import pandas as pd
from google.colab import drive
drive.mount('/content/drive')
df=pd.read_csv('/content/drive/MyDrive/iris.csv')
df.info()
df.variety.value_counts()
df.head()
features=df.iloc[:,:-1].values
label=df.iloc[:,4].values
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))
from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))
from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

variety	count
Setosa	50
Versicolor	50
Virginica	50

dtype: int64

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

```
KNeighborsClassifier()
```

```
0.9666666666666667
1.0
array([[50,  0,  0],
       [ 0, 47,  3],
       [ 0,  1, 49]])
```

	precision	recall	f1-score	support
Setosa	1.00	1.00	1.00	50
Versicolor	0.98	0.94	0.96	50
Virginica	0.94	0.98	0.96	50
accuracy			0.97	150
macro avg	0.97	0.97	0.97	150
weighted avg	0.97	0.97	0.97	150

Result:

Thus, we have successfully implemented the KNN algorithm for a given data set.